

Challenge (Habanero)

- Q1.** A recipe for cake requires $2\frac{3}{4}$ cups of flour. If you want to make half the recipe, how many cups of flour will you need?
- (A) $1\frac{3}{8}$
(B) $1\frac{3}{4}$
(C) $1\frac{1}{2}$
(D) $1\frac{1}{4}$
(E) $1\frac{5}{8}$
- Q2.** A building is 25.5 meters tall. If a scale model of the building is 0.5 meters tall, what is the scale of the model?
- (A) 1 : 50
(B) 1 : 51
(C) 1 : 52
(D) 1 : 49
(E) 1 : 48
- Q3.** A chef needs to mix $3\frac{1}{2}$ liters of sauce with $2\frac{3}{4}$ liters of water. How many liters of mixture will the chef have in total?
- (A) $6\frac{1}{4}$
(B) $6\frac{1}{2}$
(C) $6\frac{3}{4}$
(D) $7\frac{1}{4}$
(E) $7\frac{1}{2}$
- Q4.** A carpenter has a plank that is 5.8 meters long. If the carpenter cuts 2.1 meters from the plank, what fraction of the plank remains?
- (A) $\frac{3}{5}$
(B) $\frac{4}{5}$
(C) $\frac{7}{10}$
(D) $\frac{3}{8}$
(E) $\frac{7}{8}$
- Q5.** A recipe requires 1.25 cups of oil. If you only have a $\frac{1}{4}$ cup measuring cup, how many times will you need to fill the measuring cup?
- (A) 4
(B) 5
(C) 6
(D) 8
(E) 10
- Q6.** A water tank can hold 4500 liters of water. If $2\frac{1}{4}$ percent of the tank is already filled, how many liters of water are in the tank?
- (A) 90
(B) 100
(C) 101.25
(D) 112.5
(E) 101
- Q7.** A architect designed a building with a roof that is 4.2 meters above the floor. If a 2.5 meter ladder is placed against the wall, what fraction of the ladder is above the roof?
- (A) $\frac{1}{8}$
(B) $\frac{1}{7}$
(C) $\frac{3}{14}$
(D) $\frac{1}{4}$
(E) $\frac{3}{8}$
- Q8.** A painter needs to mix 2.8 liters of paint with 1.2 liters of solvent. If the painter adds 0.5 liters of water, what is the total amount of mixture?
- (A) 4.5
(B) 4.3
(C) 4.1
(D) 4.7
(E) 5.5

Open Ended Questions (FRQ)

- Q9.** A bridge is 235.6 meters long. If a section of the bridge is $3\frac{1}{2}$ meters above the water level, and the water level rises by 0.8 meters, what fraction of the bridge is above the water level? Show your work and provide your answer as a fraction and a decimal.
- Q10.** A contractor needs to pour 15.2 cubic meters of concrete into a foundation. If the concrete is 2.5 percent water, how many liters of water will be in the concrete? Show your work and provide your answer in liters.

Chef's Tips

- Check units of measurement
- Use a calculator to verify answers
- Emphasize showing work and explanations